Accelerating Polynomial System Solving via Factorization and Unsatisfiable-Core Learning

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Abstract

Polynomial systems are a fundamental language for describing algebraic and geometric problems across mathematics and its applications. Beyond finding individual solutions, one often seeks global properties of the solution space, such as its dimension and irreducible components, or structural properties and decompositions of the underlying polynomial ideal. For large polynomial systems with many variables and high degrees, determining these global properties can become intractable. Leading approaches include variants of Buchberger's algorithm with syzygy, decomposition techniques, resultants, and cylindrical algebraic decomposition. However, even the most modern approaches still often fail on practical problems or slight perturbations of standard benchmarks. Here we introduce a framework that divides a polynomial system into simpler branches. It assumes only that the branch solver describes satisfiable branches and certifies unsatisfiable (UNSAT) ones. If the solver also returns an UNSAT core, the framework uses it to prune later branches, following the same basic idea as conflict-driven clause learning (CDCL). In our experiments, we use Gröbner-basis solvers and apply the method to two systems arising from research in quantum field theory for which no standard method in our comparison succeeds. At a later stage, we construct a decision tree guided by domain-informed hypotheses to efficiently enumerate candidate solutions. Together, these ideas offer a path toward the exact analysis of previously inaccessible polynomial systems.

1 Introduction

Systems of multivariate polynomial equations appear whenever a symbolic model contains nonlinear algebraic constraints. Applications include robotics, coding the-

ory, optimization, statistics, and physics [11]. In many of these problems, an isolated solution is not enough. One needs the dimension of the solution set, its irreducible components, or exact relations that hold on every solution. These are questions about the variety and its defining ideal, rather than about a single point [2, 7].

Elimination by resultants provided an early systematic way to attack polynomial systems, but the introduction of Gröbner bases gave a common algorithmic framework for elimination, ideal membership, dimension, and related tasks. Buchberger developed this framework in his 1965 thesis [1]. Later algorithms changed how the expensive reductions are organized. F4 replaces individual polynomial reductions with sparse linear algebra, while F5 uses signatures to avoid many reductions to zero [3, 4]. Primary-decomposition algorithms use Gröbner bases and factorization to separate components [5]. Numerical algebraic geometry takes a different route: homotopy continuation and witness sets provide isolated roots numerically [9, 10].

Recent learning-based work has concentrated on these choices. Peifer et al. trained a reinforcement-learning policy for pair selection in Buchberger's algorithm [8]. Kera et al. trained transformers to compute Gröbner bases for a restricted class of zero-dimensional radical ideals [6]. Both results concern controlled input distributions; a large, general ideal still requires an exact symbolic computation.

Our method starts with that factor structure. Many equations in our target systems factor. If $p_i = \prod_j f_{ij}$, then every solution of $p_i = 0$ makes at least one factor f_{ij} vanish. Selecting factors that meet every equation gives a hitting set of the factor family. It is enough to enumerate the minimal hitting sets, and each one defines a smaller polynomial system. We solve these systems separately. The union of their zero sets is exactly the zero set of the original input, so the split does not discard solutions or rely on a genericity assumption.

The framework uses UNSAT core learning to prune hitting sets and the accelerate computation. The method does not require a particular branch solver or core-extraction procedure.

Some factor branches can still be too expensive for the chosen subsolver. After a timeout, we branch again splitting a selected variable into the cases $x = 0$ and $x \neq 0$. A domain expert may have a good reason to choose a different splitting which will yield faster computation.

We prove that our methods preserve the complete solution set for any branch solver satisfying the stated conditions. We compare the resulting implementation with direct Gröbner-basis computation, symbolic decomposition, and numerical homotopy on generated benchmarks. We then apply it to two systems from quantum field theory for which the standard methods in our comparison do not complete. The output is an exact collection of branch descriptions for the full solution set, including

positive-dimensional families.

2 Preliminaries and Notation

2.1 Notation

Let K be a field, let $\overline{K}$ be an algebraic closure of K, and let $K[x_1, \ldots, x_n]$ be the polynomial ring in n variables over K. We write the input system as

$$P = \{p_1, \ldots, p_m\} \subseteq K[x_1, \ldots, x_n].$$

We assume that each p_i is nonzero and fix a factorization

$$p_i = a_i \prod_{f \in C_i} f^{e_{i,f}},$$

where $a_i \in K \setminus \{0\}$, C_i is the set of distinct nonconstant factors appearing in the factorization, and $e_{i,f} \geq 1$ is the multiplicity of f.

For any set of polynomials $Q \subseteq K[x_1, \ldots, x_n]$, we write $\langle Q \rangle$ for the ideal generated by Q and define its zero set over the algebraic closure by

$$V(Q) = \{x \in \overline{K}^n : q(x) = 0 \text{ for every } q \in Q\}.$$

For $X \subseteq \overline{K}^n$, its vanishing ideal is

$$I(X) = \{g \in K[x_1, \ldots, x_n] : g(x) = 0 \text{ for every } x \in X\}.$$

Unless stated otherwise, satisfiability refers to solutions over $\overline{K}$. Hilbert's Nullstellensatz [2] gives

$$V(Q) = \emptyset \quad \Longleftrightarrow \quad \langle Q \rangle = \langle 1 \rangle.$$

Equivalently, the reduced Gröbner basis of $\langle Q \rangle$ is $\{1\}$. A set $U \subseteq Q$ is an UNSAT core of Q if $V(U) = \emptyset$.

A hitting set of $C_1, \ldots, C_m$ is a set $H \subseteq \bigcup_{i=1}^{m} C_i$ such that $H \cap C_i \neq \emptyset$ for every i. It is minimal if none of its proper subsets is also a hitting set. We denote the collection of all minimal hitting sets of $C_1, \ldots, C_m$ by $\mathcal{H}$.

2.2 Polynomial ideals

2.3 Basic algebraic geometry

3 Method

Algorithm 1 uses a branch solver as a subroutine. Given a polynomial system P, it factors each polynomial to obtain $C_1, \ldots, C_m$ and enumerates the minimal hitting sets of these factor sets. Each hitting set is itself a smaller polynomial system. For a hitting set H, the branch solver must either describe $V(H)$ exactly or certify that $V(H) = \emptyset$. It may also return a certified UNSAT core $U \subseteq H$; any later hitting set containing U can be skipped.

The branch solver returns a representation of the solution set of each satisfiable branch. If S is a collection of such representations, let $\mathrm{Sol}(S)$ denote the union of the represented sets. The algorithm returns S with $\mathrm{Sol}(S) = V(P)$. It processes the hitting sets in estimated order of cost and skips any set that contains a learned core.

Algorithm 1 Branching

Require: A set of polynomials $P = \{p_1, \ldots, p_m\}$ and an exact branch solver $\mathcal{B}$.

Interreduce the polynomials ▷ Optional
Initialize $C_1, \ldots, C_m$ as the sets of factors of $p_1, \ldots, p_m$
Initialize $C \leftarrow \emptyset$ an UNSAT-core set
Initialize $S \leftarrow \emptyset$ a set of solution-set representations
for Each minimal hitting set H of $C_1, \ldots, C_m$ **do**
    if H contains an UNSAT core from C **then**
        continue
    end if
    Use $\mathcal{B}$ to solve H
    if $\mathcal{B}$ returns a representation of $V(H)$ **then**
        Add the returned representation to S
    else
        if $\mathcal{B}$ returns a certified UNSAT core **then**
            Add the core to C
        end if
    end if
end for
return S, C

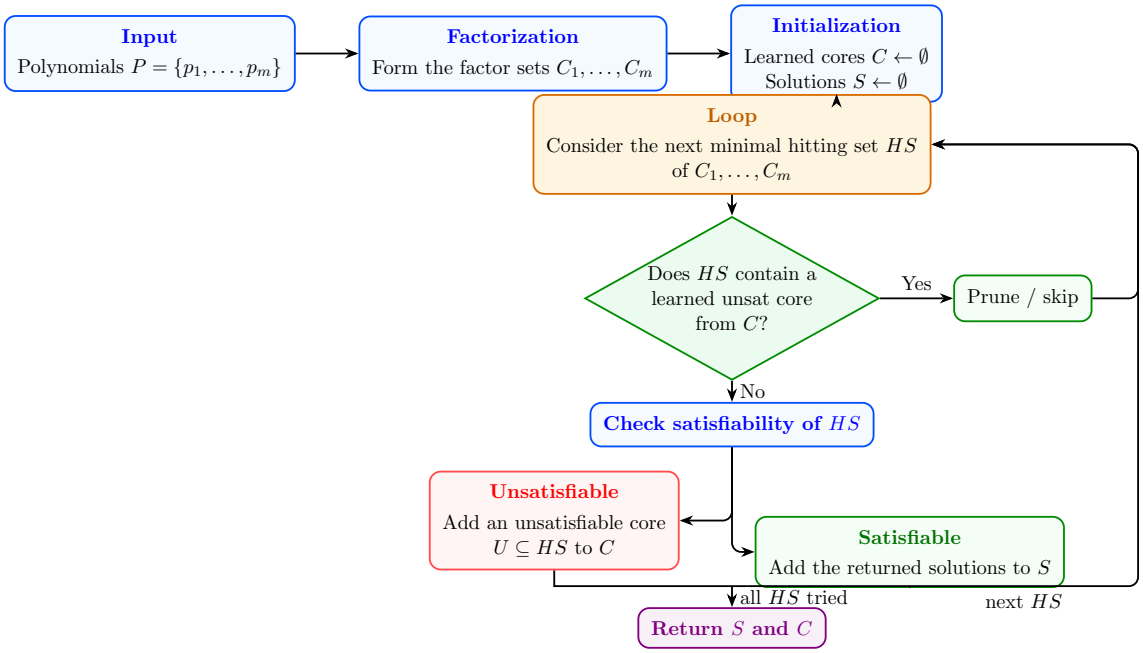


Figure 1: Flowchart of the branching procedure of Algorithm 1.

The order of the minimal hitting sets affects running time but not correctness. We use the number of factors in a hitting set as an estimate of its cost and process smaller sets first. An UNSAT core found in a small set may then prune larger sets before they reach the solver.

3.1 Theoretical analysis

Lemma 1 (Correctness of interreduction) *Suppose that one generator p_i is replaced by*

$$p_i' = p_i - \sum_{j \neq i} a_j p_j, \qquad a_j \in K[x_1, \ldots, x_n].$$

Then

$$\langle p_1, \ldots, p_i, \ldots, p_m \rangle = \langle p_1, \ldots, p_i', \ldots, p_m \rangle.$$

Any finite sequence of these reductions therefore preserves the generated ideal and its zero set.

Proof 1 *The polynomial p_i' belongs to the original ideal, while $p_i = p_i' + \sum_{j \neq i} a_j p_j$ belongs to the ideal generated by the updated system. Each generating set therefore*

generates the other. The same argument applies at every step of a sequential interreduction.

A zero generator imposes no constraint and can be discarded. A nonzero constant makes the ideal the unit ideal, so the input system is UNSAT.

Lemma 2 (Sufficiency of minimal hitting sets) *Every hitting set T of the factor family contains a minimal hitting set $H \subseteq T$, with $V(T) \subseteq V(H)$. It is therefore enough to consider minimal hitting sets.*

Proof 2 *Because the factor sets are finite, repeatedly removing redundant factors from T eventually produces a minimal hitting set H. Since $H \subseteq T$, any point in $V(T)$ also lies in $V(H)$.*

Theorem 1 *The solution set of P is*

$$V(P) = \bigcup_{H \in \mathcal{H}} V(H).$$

Proof 3 *Take $x \in V(P)$. Since $p_i(x) = 0$, at least one factor in C_i vanishes at x for every i. The set of all factors that vanish at x is therefore a hitting set of $C_1, \ldots, C_m$. Because the factor sets are finite, this set contains a minimal hitting set H. Every factor in H vanishes at x, so $x \in V(H)$.*

For the reverse inclusion, take $x \in V(H)$ for some $H \in \mathcal{H}$. The set H intersects every C_i, so each p_i has a factor that vanishes at x. Hence $p_i(x) = 0$ for every i, and $x \in V(P)$.

The same decomposition can be stated in terms of ideals.

Theorem 2 (Radical decomposition) *The radical of the input ideal satisfies*

$$\sqrt{\langle P \rangle} = \bigcap_{H \in \mathcal{H}} \sqrt{\langle H \rangle}.$$

Proof 4 *By Theorem 1, $V(P) = \bigcup_{H \in \mathcal{H}} V(H)$. Taking vanishing ideals gives*

$$I(V(P)) = I\left(\bigcup_{H \in \mathcal{H}} V(H) \right) = \bigcap_{H \in \mathcal{H}} I(V(H)).$$

Hilbert's Nullstellensatz identifies $I(V(P))$ with $\sqrt{\langle P \rangle}$ and each $I(V(H))$ with $\sqrt{\langle H \rangle}$, which gives the stated equality.

Lemma 3 *Let U be an UNSAT core. If $U \subseteq H$, then $V(H) = \emptyset$.*

Proof 5 *Since every point in $V(H)$ must satisfy every polynomial in U, we have $V(H) \subseteq V(U)$. The core U is UNSAT, so $V(U) = \emptyset$ and therefore $V(H) = \emptyset$.*

Soundness does not require U to be minimal. A smaller core is useful only because more later branches may contain it.

For the following result, assume that the branch solver terminates on every unpruned branch H. It either returns $V(H)$ exactly or certifies that $V(H) = \emptyset$. Any core used for pruning must be a certified subset $U \subseteq H$.

Theorem 3 *Algorithm 1 is sound and complete for P.*

Proof 6 *Let P' denote the system after the optional interreduction step. We check that each part of the algorithm preserves the solution set.*

Interreduction. *Lemma 1 gives $\langle P' \rangle = \langle P \rangle$, and hence $V(P') = V(P)$.*

Factorization and branch construction. *Theorem 1, applied to the factor sets of P', gives*

$$V(P') = \bigcup_{H \in \mathcal{H}} V(H).$$

Lemma 2 shows that no nonminimal hitting set can add a solution missing from this union. Thus the initial branch collection represents exactly $V(P')$.

Loop invariant. *Let R be the set of unprocessed branches and S the collection of solution-set representations returned so far. At the start of the loop,*

$$\mathrm{Sol}(S) \subseteq V(P') \qquad \textit{and} \qquad V(P') \subseteq \mathrm{Sol}(S) \cup \bigcup_{H \in R} V(H).$$

The first inclusion is the soundness invariant. The second is the completeness invariant: every solution outside $\mathrm{Sol}(S)$ remains in a pending branch. Both hold initially because $S = \emptyset$, $R = \mathcal{H}$, and the branch decomposition holds.

Core pruning. *Suppose the algorithm skips H because it contains a learned core U. Lemma 3 gives $V(H) = \emptyset$. Removing H from R therefore leaves the completeness invariant unchanged, so soundness is unchanged as well.*

Branch solving. *If H is satisfiable, the solver adds to S a representation of $V(H)$. Since $V(H) \subseteq V(P')$, the soundness invariant is preserved. Moving $V(H)$ from the pending union into* $\mathrm{Sol}(S)$ *also preserves the completeness invariant. If H is UNSAT, then $V(H) = \emptyset$, so removing it from R preserves both invariants.*

Return. *When the loop ends, $R = \emptyset$. The completeness invariant gives $V(P') \subseteq \mathrm{Sol}(S)$, while the soundness invariant gives* $\mathrm{Sol}(S) \subseteq V(P')$. *Hence* $\mathrm{Sol}(S) = V(P') = V(P)$.

The theoretical analysis does not require K to be algebraically closed. If a system has no solution over the algebraic closure, it has no solution over K.

4 Branching with variable zeroing

Algorithm 2 handles timeouts from the branch solver. After a timeout, it chooses a variable x and replaces the unresolved branch with two children, one with the additional constraint $x = 0$ and another with $x \neq 0$. It then applies Algorithm 1 recursively to both children. The recursion stops when all branches are resolved, no splitting variables remain, or the computation budget is exhausted. This construction does not depend on the choice of branch solver.

Algorithm 2 Recursive branching with variable zeroing

Require: A polynomial system P, and a timeout τ.

function ZeroingBranching(P, τ)
  $C \leftarrow \emptyset$ ▷ One global UNSAT-core set
  $X \leftarrow$ Variables(P) ▷ All variables of P
  return SolveByZeroing(P, X, τ, C)
end function
function SolveByZeroing(P, X, τ, C)
  $(S, T) \leftarrow$ BranchingWithTimeout(P, τ, C) ▷ Algorithm 1; C is shared and updated
  $U \leftarrow \emptyset$ ▷ Branches left unresolved
  for all $Q \in T$ **do**
    if $X = \emptyset$ **then**
      $U \leftarrow U \cup \{Q\}$
      continue
    end if
    $x \leftarrow$ ChooseVariable(Q, X)
    $(S_0, U_0, C) \leftarrow$ SolveByZeroing($Q \cup \{x\}, X \setminus \{x\}, \tau, C$)
    $(S_1, U_1, C) \leftarrow$ SolveByZeroing($Q \cup \{1 - xz\}, X \setminus \{x\}, \tau, C$) ▷ z is fresh
    $S \leftarrow S \cup S_0 \cup S_1$
    $U \leftarrow U \cup U_0 \cup U_1$
  end for
  return (S, U, C)
end function

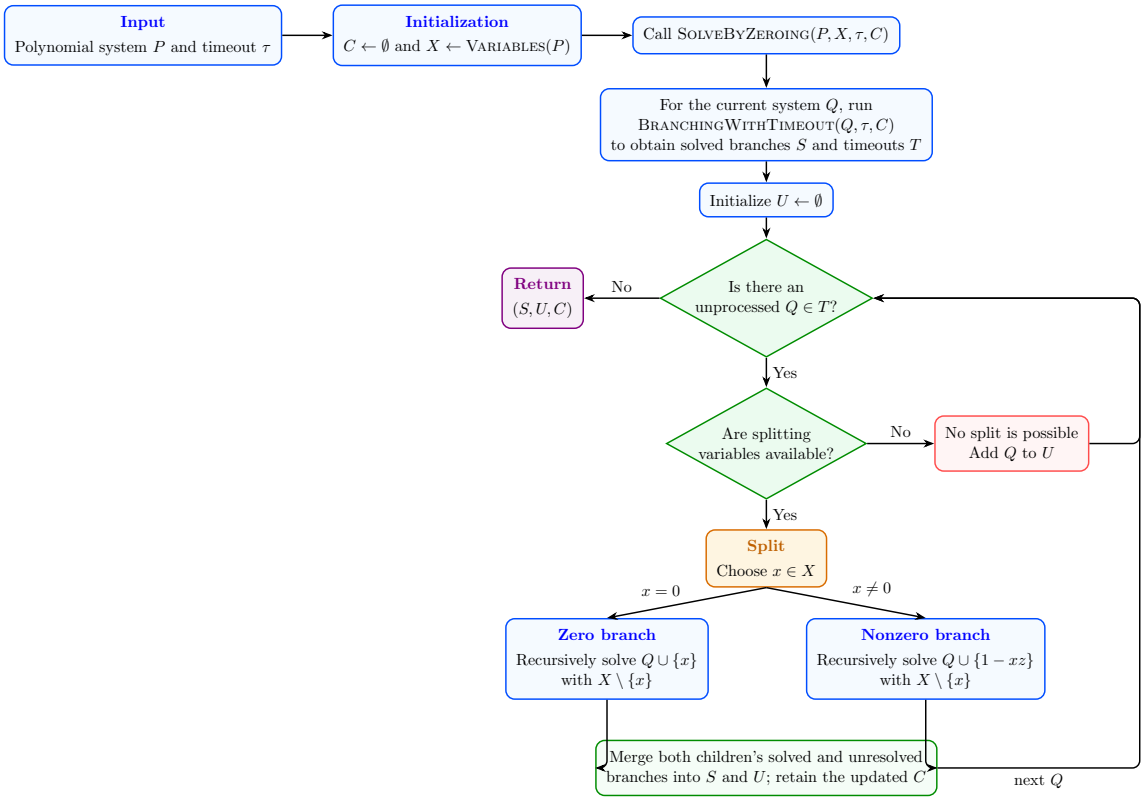


Figure 2: Flowchart of the recursive variable-zeroing procedure of Algorithm 2.

Solutions of the nonzero branch are projected onto the original variables, so the fresh variables do not appear in the output.

In the zero branch, the solver can substitute $x = 0$ directly. In the nonzero branch, $1 - xz = 0$ records that x is invertible. Interreduction with either equation may simplify the remaining generators. The effect depends on the branch solver. Section 5 describes the Gröbner-basis solvers used in our experiments.

The run on `systems/system.txt`, with a five-second timeout for each hitting set, produces the recursion tree in Figure 3. The variable chosen after a timeout is written inside the corresponding hitting-set node. Its outgoing edges give the zero and nonzero cases.

4.1 Choice of variable for splitting

The choice of variable for splitting effects the efficiency of the algorithm. Every split creates at least two new hitting sets to check. If a variable barely occurs in

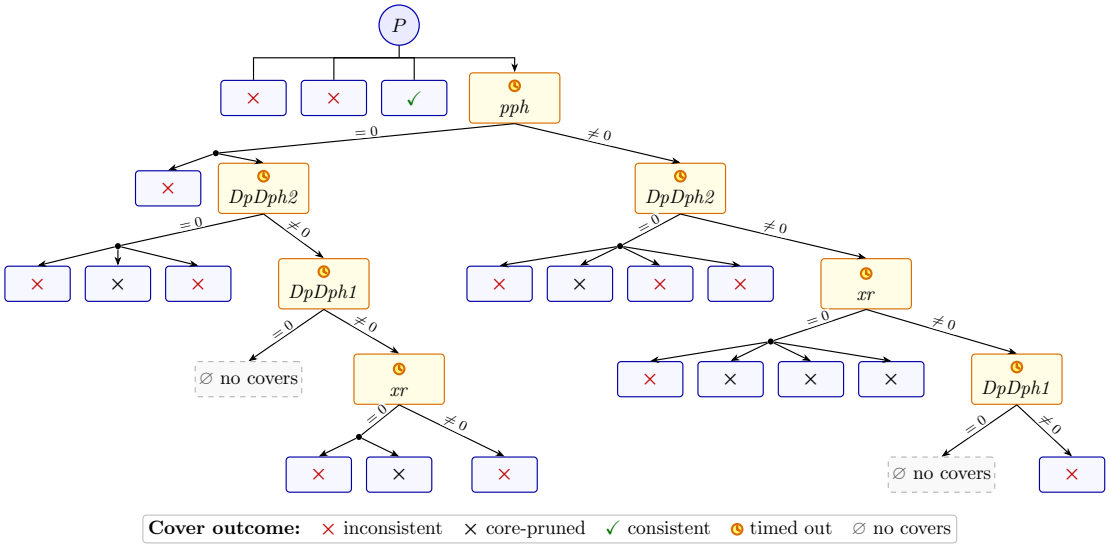


Figure 3: Recursion tree for `systems/system.txt`. Each hitting set is shown separately; ∅ marks a branch that produces no hitting sets.

the current hitting set, both recursive calls may repeat most of the same work. A variable that occurs in many generators is a more promising choice: setting it to zero removes all terms containing it at once, while in the nonzero branch the algorithm can use the $1 - xz$.

For each candidate variable x that is still available on the current branch, define

$$\begin{aligned} g_Q(x) &= \#\{q \in Q : x \text{ occurs in } q\}, \\ t_Q(x) &= \#\{(q, m) : q \in Q,\ m \text{ is a term of } q \deg_x(m) > 0\}, \\ e_Q(x) &= \sum_{q \in Q} \sum_{m \text{ a term of } q \deg_x(m) > 0} \deg_x(m). \end{aligned}$$

Its score is the triple

$$h_Q(x) = \big(g_Q(x), t_Q(x), e_Q(x)\big),$$

and scores are compared lexicographically.

4.2 Theoretical analysis of variable zeroing

Let Y be the set of variables in the input system, and let π_Y denote projection onto those coordinates. For any system Q encountered during the recursion, write

$$V_Y(Q) = \pi_Y(V(Q)).$$

This projection removes the inverse variables introduced along the path to Q.

We use the following contract for BranchingWithTimeout. On input Q, it returns a collection S_Q of solution-set representations and a set T_Q of unresolved branches such that

$$V_Y(Q) = \mathrm{Sol}(S_Q) \cup \bigcup_{R \in T_Q} V_Y(R). \tag{1}$$

As defined in Section 3, $\mathrm{Sol}(S_Q)$ is the union of the sets represented by S_Q, projected here onto the coordinates in Y. Equation (1) is the usual branching invariant: a timeout moves an unprocessed branch to T_Q instead of discarding it. We also assume, as in Theorem 3, that every learned UNSAT core is certified.

Lemma 4 (Correctness of a zeroing split) *Let $x \in Y$, and let z be a variable that does not occur in Q. Then*

$$V_Y(Q) = V_Y(Q \cup \{x\}) \cup V_Y(Q \cup \{1 - xz\}).$$

Proof 7 *Take $a \in V_Y(Q)$ and choose a point $\widetilde{a} \in V(Q)$ with $\pi_Y(\widetilde{a}) = a$. If $\widetilde{a}_x = 0$, then $a \in V_Y(Q \cup \{x\})$. If $\widetilde{a}_x \neq 0$, extend $\widetilde{a}$ by setting $z = \widetilde{a}_x^{-1}$. The resulting point satisfies $1 - xz = 0$, so $a \in V_Y(Q \cup \{1 - xz\})$. This proves the forward inclusion. The reverse inclusion follows because each child system contains every polynomial in Q.*

Theorem 4 (Variable zeroing) *Suppose Algorithm 2 returns (S, U, C) on input P. Then*

$$V(P) = \mathrm{Sol}(S) \cup \bigcup_{R \in U} V_Y(R).$$

In particular, if $U = \emptyset$, then $\mathrm{Sol}(S) = V(P)$.

Proof 8 *We prove the stronger claim that every call* SolveByZeroing(Q, X, τ, C) *returning (S', U', C') satisfies*

$$V_Y(Q) = \mathrm{Sol}(S') \cup \bigcup_{R \in U'} V_Y(R). \tag{2}$$

The proof is by induction on $|X|$.

If $X = \emptyset$, the procedure cannot split an unresolved branch. It returns the representations found by BranchingWithTimeout *and places every branch in T_Q into U'. Thus, Equation* (2) *is exactly Equation* (1)*.*

Now suppose $X \neq \emptyset$ and that the claim holds whenever fewer variables are available. Let (S_Q, T_Q) be the result of BranchingWithTimeout *on Q. For each $R \in T_Q$, the procedure chooses some $x \in X$ and recursively processes*

$$R_0 = R \cup \{x\} \qquad \text{and} \qquad R_1 = R \cup \{1 - xz\},$$

where z is fresh. Both calls use the smaller set $X \setminus \{x\}$. Lemma 4 gives

$$V_Y(R) = V_Y(R_0) \cup V_Y(R_1).$$

Substituting these decompositions into Equation (1) *gives Equation* (2).

Apply the claim to the initial call with $Q = P$. Since P contains no inverse variables, $V_Y(P) = V(P)$, which proves the stated identity. If U is empty, the union over U is empty and hence $\mathrm{Sol}(S) = V(P)$.

Now to show the run always terminates in finite time.

Theorem 5 *Zeroing-Branching 2 terminates in finite time.*

Proof 9 *The depth of the recursion tree is at most $|X|$ since for each call an unused variable is required and removed. The branching factor is bounded by the number of hitting sets of factors which is finite. The tree size is finite, and due to the timeout each hitting set is evaluated in finite time. Thus the algorithm terminates in finite time.*

We can infer a solution-set representation for the hitting set on which the solver timed out from its children's representations. If all children are UNSAT, then so is the result of the timeout evaluation. Otherwise it represents the union of the sets represented by its children.

5 Experiments

5.1 Setup

We implemented Algorithm 1 in Rust 1.97.1 and used Symbolica 1.5 for polynomial arithmetic over the rationals. In all experiments, we solve the branches with a Gröbner-basis computation. The available implementations are M4GB, F5, GVW, F4, and Singular 4.4.1. A basis containing 1 certifies that the branch is UNSAT. We obtain an UNSAT core by iterative refinement as in Section 5 of [12]. The comparison also includes six direct whole-ideal Gröbner-basis configurations, three

decomposition methods from Singular, Macaulay2 1.26.06, and a numerical method from HomotopyContinuation.jl under Julia 1.12.6.

For Algorithm 2, we test M4GB, F5, GVW, Symbolica, and Singular as branch solvers, disable UNSAT-core learning, and impose a three-second limit on each cover computation. We selected this cutoff by trying one, two, three, and five seconds on the hard gated systems. Three seconds is the smallest tested value that lets Singular complete the Katsura-8 residual (about 2.6 seconds); five seconds completes no additional hard system and only delays the subsequent splits.

We ran the benchmark on an Apple M4 Pro with 14 CPU cores and 24 GB of memory. Every table entry is a single run. Every method received 60 seconds per system; the three-second zeroing limit is an inner, per-cover limit within that budget.

For branching and variable zeroing, T is end-to-end wall time and F is the time from the start of the minimal-hitting-set search to its first satisfiable hitting set. Factorization and input processing are therefore included in T but not in F. The complete measurements are stored in the appendix.

5.2 Baseline systems

The baseline has 20 generated systems in six families. Signed-distance cycles and affine-factor hypergraphs test many inexpensive hitting sets. The overlapping random multiples use seeded dense quadratic factors that recur across equations, with no factor shared by every equation. In the gated systems, a single selector multiplies every equation of a Cyclic, Katsura, or dense quadratic system. This produces one easy selector cover and one nonlinear residual cover.

Family	Tested sizes	Variables/equations; hitting sets
Signed-distance cycle	$n = 8, 10, 12$	$8/8, 10/10, 12/12$; $8, 8, 32$
Overlapping random multiples	$(n, p, k) = (4, 6, 2), (6, 8, 3), (8, 10, 3)$	n/n; $4, 16, 28$
Positive affine hypergraph	$(n, m, k) = (8, 6, 4), (10, 8, 5), (12, 10, 6)$	n/m; $63, 239, 833$
Gated Cyclic	orders $5, 6, 7$	$6/5, 7/6, 8/7$; 2 each
Gated Katsura and dense quadratics	Katsura $6, 7, 8$; dense $n = 5, 6, 7$	$8/7, 9/8, 10/9, 6/5, 7/6, 8/7$; 2 each
Square affine hypergraph	$(n, k) = (8, 4), (10, 5)$	n/n; $31, 304$

Table 1: The generated systems used for the baseline benchmark. The overlapping random multiples use seed 31415; both affine families use seed 42.

The overlapping-random-multiple runs disable interreduction so that the generated factor-incidence pattern reaches the hitting-set enumerator unchanged. All other branching runs use interreduction. Table 2 selects the fastest completed branching backend, the fastest completed zeroing backend observed, and the fastest completed method in each baseline class. A branching or zeroing entry gives T/F in

seconds; the other entries give total wall time. "Timeout" means that every tested method in that class timed out. "Partial" means that zeroing returned at least one unresolved branch, and "error" records a failed run rather than a mathematical conclusion.

System	Best branching T/F	Selected zeroing T/F	Best direct GB	Best decomposition	Numerical
Signed distance, $n = 8$	Symb. 0.008/0.000	Symb. 0.008/0.001	Symb. 0.008	facstd 0.190	32.410
Signed distance, $n = 10$	Symb. 0.025/0.002	Symb. 0.025/0.002	Symb. 0.029	facstd 0.430	29.944
Signed distance, $n = 12$	Symb. 0.087/0.002	Symb. 0.091/0.003	Symb. 0.472	primdec 3.056	36.880
Overlapping $(4, 6, 2)$	F5 0.008/0.000	F5 0.009/0.001	Symb. 0.413	facstd 0.265	30.069
Overlapping $(6, 8, 3)$	F5 0.132/0.001	F5 0.132/0.021	timeout	timeout	timeout
Overlapping $(8, 10, 3)$	Singular 2.931/0.010	Singular 2.892/0.152	timeout	timeout	timeout
Positive affine $(8, 6, 4)$	F5 0.008/0.000	Singular 0.181/0.012	timeout	facstd 0.368	50.934
Positive affine $(10, 8, 5)$	F5 0.020/0.000	Symb. 0.060/0.007	timeout	facstd 1.836	timeout
Positive affine $(12, 10, 6)$	F5 0.067/0.000	Singular 9.600/0.078	timeout	facstd 53.318	timeout
Gated Cyclic-5	Symb. 0.009/0.000	Symb. 0.011/0.001	Symb. 0.008	facstd 0.181	35.054
Gated Cyclic-6	Singular 0.045/0.008	Singular 0.043/0.009	Singular 0.309	M2 2.620	37.314
Gated Cyclic-7	timeout	Singular partial 24.118/0.009	timeout	timeout	58.287
Gated Katsura-6	Singular 0.049/0.008	Singular 0.049/0.009	GVW 0.226	facstd 0.233	31.532
Gated Katsura-7	Singular 0.283/0.008	Singular 0.284/0.009	GVW 3.176	facstd 0.539	31.980
Gated Katsura-8	Singular 2.657/0.052	Singular 2.588/0.009	timeout	facstd 3.596	34.264
Gated dense, $n = 5$	Singular 0.029/0.008	Singular 0.029/0.008	M4GB 0.171	facstd 0.198	29.289
Gated dense, $n = 6$	Singular 0.546/0.008	Singular 0.535/0.008	M4GB 4.736	facstd 0.755	30.053
Gated dense, $n = 7$	Singular 42.020/0.017	error	timeout	facstd 48.820	32.261
Square affine $(8, 4)$	Singular 0.470/0.023	Singular 0.445/0.012	timeout	facstd 0.601	timeout
Square affine $(10, 5)$	Symb. 1.174/0.000	Symb. 1.326/0.015	timeout	facstd 2.141	timeout

Table 2: Best completed baseline results and the fastest completed zeroing backend observed. Branching and zeroing entries show wall time and first-solution latency; baseline entries show wall time. All times are in seconds.

5.3 Physical Systems

6 Conclusion

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A Complete experimental results

The tables below give every run summarized in Table 2. Unless stated otherwise, entries are end-to-end wall times in seconds. P marks a partial run and TO an outer

timeout. These are single measurements.

A.1 Factor branching

Table 3 reports all five branch-solver configurations. A partial entry found the easy component shown in the main text but left at least one cover unresolved.

A.2 Variable zeroing

Table 4 gives the complete output of the fastest completed zeroing configuration used in Table 2. Every backend receives three seconds per cover and a 60-second outer budget, and UNSAT-core learning is disabled. The overlapping-random-multiple rows disable interreduction, as in the ordinary branching experiment. Here N is the number of recursion nodes, E the number of cover evaluations, O/S the numbers of per-cover timeouts and zeroing splits, and C/U the numbers of returned components and unresolved branches. The time entry is T/F.

The dense $n = 7$ run failed while constructing the first recursive child: the expanded Singular program exceeded the operating system's command-line argument limit. This is an implementation error, not an UNSAT result or a timeout. We also tried M4GB, F5, GVW, and Symbolica on this system. These backends execute in-process; after a per-cover timeout their non-cancellable worker threads continued alongside the recursive children until the process was killed under memory pressure. Singular remains preferable for the hard rows because its external worker can be terminated. The Cyclic-7 run exhausted all useful splitting variables on one route and therefore returned one unresolved branch.

A.3 Direct whole-ideal Gröbner bases

A.4 Decomposition and numerical methods

System	M4GB	F5	GVW	Symbolica	Singular
Signed distance, $n = 8$	0.651	0.053	0.122	0.008	0.083
Signed distance, $n = 10$	0.046	10.927	45.797	0.025	0.084
Signed distance, $n = 12$	0.182	6.256	1.841	0.087	0.271
Overlapping $(4, 6, 2)$	0.017	0.008	0.011	0.015	0.038
Overlapping $(6, 8, 3)$	1.053	0.132	0.280	0.657	0.201
Overlapping $(8, 10, 3)$	TO	10.937	30.687	TO	2.931
Positive affine $(8, 6, 4)$	0.009	0.008	0.010	0.008	0.497
Positive affine $(10, 8, 5)$	0.021	0.020	0.028	0.020	1.870
Positive affine $(12, 10, 6)$	0.072	0.067	0.096	0.068	6.498
Gated Cyclic-5	0.010	0.010	0.011	0.009	0.022
Gated Cyclic-6	P 60.010	5.960	2.275	P 60.013	0.045
Gated Cyclic-7	P 60.875	P 60.504	P 60.955	P 60.024	P 60.103
Gated Katsura-6	0.366	0.483	1.338	2.540	0.049
Gated Katsura-7	4.487	15.220	42.113	P 60.012	0.283
Gated Katsura-8	54.055	P 60.325	P 60.047	P 60.012	2.657
Gated dense, $n = 5$	0.177	0.071	0.200	0.204	0.029
Gated dense, $n = 6$	4.716	3.194	7.475	21.498	0.546
Gated dense, $n = 7$	P 60.008	P 60.050	P 60.041	P 60.015	42.020
Square affine $(8, 4)$	15.117	TO	TO	3.096	0.470
Square affine $(10, 5)$	14.750	TO	47.561	1.174	2.548

Table 3: End-to-end wall times for every factor-branching configuration.

System	Backend	T/F	N	E	O/S	C/U	Result
Signed distance, $n = 8$	Symbolica	0.008/0.001	1	8	0/0	8/0	complete
Signed distance, $n = 10$	Symbolica	0.025/0.002	1	8	0/0	8/0	complete
Signed distance, $n = 12$	Symbolica	0.091/0.003	1	32	0/0	32/0	complete
Overlapping $(4, 6, 2)$	F5	0.009/0.001	1	4	0/0	4/0	complete
Overlapping $(6, 8, 3)$	F5	0.132/0.021	1	16	0/0	16/0	complete
Overlapping $(8, 10, 3)$	Singular	2.892/0.152	1	28	0/0	28/0	complete
Positive affine $(8, 6, 4)$	Singular	0.181/0.012	1	4	0/0	4/0	complete
Positive affine $(10, 8, 5)$	Symbolica	0.060/0.007	1	242	0/0	242/0	complete
Positive affine $(12, 10, 6)$	Singular	9.600/0.078	1	511	0/0	511/0	complete
Gated Cyclic-5	Symbolica	0.011/0.001	1	2	0/0	2/0	complete
Gated Cyclic-6	Singular	0.043/0.009	1	2	0/0	2/0	complete
Gated Cyclic-7	Singular	24.118/0.009	15	17	8/7	1/1	partial
Gated Katsura-6	Singular	0.049/0.009	1	2	0/0	2/0	complete
Gated Katsura-7	Singular	0.284/0.009	1	2	0/0	2/0	complete
Gated Katsura-8	Singular	2.588/0.009	1	2	0/0	2/0	complete
Gated dense, $n = 5$	Singular	0.029/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 6$	Singular	0.535/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 7$	Singular	4.797/–	–	–	–	–	error
Square affine $(8, 4)$	Singular	0.445/0.012	1	31	0/0	31/0	complete
Square affine $(10, 5)$	Symbolica	1.326/0.015	1	304	0/0	304/0	complete

Table 4: Selected variable-zeroing result for every system. Times are end-to-end wall time and first-solution latency in seconds.

System	M4GB	F5	GVW	Symbolica	Singular	Macaulay2
Signed distance, $n = 8$	0.646	0.014	0.010	0.008	0.333	0.570
Signed distance, $n = 10$	0.130	0.221	0.112	0.029	0.194	0.518
Signed distance, $n = 12$	3.266	20.044	8.685	0.472	0.512	1.293
Overlapping $(4, 6, 2)$	1.000	24.865	0.472	0.413	TO	TO
Overlapping $(6, 8, 3)$	TO	TO	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	TO	TO	TO	TO	TO	TO
Positive affine $(10, 8, 5)$	TO	TO	TO	TO	TO	TO
Positive affine $(12, 10, 6)$	TO	TO	TO	TO	TO	TO
Gated Cyclic-5	0.012	0.279	0.009	0.008	0.183	0.480
Gated Cyclic-6	TO	TO	1.738	TO	0.309	0.917
Gated Cyclic-7	TO	TO	TO	TO	TO	TO
Gated Katsura-6	0.436	44.961	0.226	5.277	14.430	TO
Gated Katsura-7	5.165	TO	3.176	TO	TO	TO
Gated Katsura-8	TO	TO	TO	TO	TO	TO
Gated dense, $n = 5$	0.171	8.884	0.179	0.200	0.529	TO
Gated dense, $n = 6$	4.736	TO	6.755	17.472	TO	TO
Gated dense, $n = 7$	TO	TO	TO	TO	TO	TO
Square affine $(8, 4)$	TO	TO	TO	TO	TO	TO
Square affine $(10, 5)$	TO	TO	TO	TO	TO	TO

Table 5: End-to-end wall times for every direct whole-ideal Gröbner-basis configuration.

System	facstd	primdecGTZ	Macaulay2	Homotopy
Signed distance, $n = 8$	0.190	0.320	1.020	32.410
Signed distance, $n = 10$	0.430	0.667	3.224	29.944
Signed distance, $n = 12$	4.800	3.056	25.223	36.880
Overlapping $(4, 6, 2)$	0.265	TO	TO	30.069
Overlapping $(6, 8, 3)$	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	0.368	TO	TO	50.934
Positive affine $(10, 8, 5)$	1.836	TO	TO	TO
Positive affine $(12, 10, 6)$	53.318	TO	TO	TO
Gated Cyclic-5	0.181	0.337	0.709	35.054
Gated Cyclic-6	2.729	TO	2.620	37.314
Gated Cyclic-7	TO	TO	TO	58.287
Gated Katsura-6	0.233	TO	TO	31.532
Gated Katsura-7	0.539	TO	TO	31.980
Gated Katsura-8	3.596	TO	TO	34.264
Gated dense, $n = 5$	0.198	TO	TO	29.289
Gated dense, $n = 6$	0.755	TO	TO	30.053
Gated dense, $n = 7$	48.820	TO	TO	32.261
Square affine $(8, 4)$	0.601	TO	TO	TO
Square affine $(10, 5)$	2.141	TO	TO	TO

Table 6: End-to-end wall times for every decomposition and numerical configuration.